

DAWBalloon: An Intuitive Musical Interface Using Floating Balloons

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ABSTRACT

The development of music synthesis technology has created a way to enjoy listening to music and actively manipulate it. However, it is difficult for an amateur to combine sounds or change pitches to operate a DAW (Digital Audio Workstation). Therefore, we focused on ultrasonic levitation and haptic feedback to develop an appropriate interface for DAW. We propose "DAWBalloon", a system that uses ultrasonic levitation arrays to visualize rhythms using floating balloons as a metaphor for musical elements and to combine sounds by manipulating the balloons. DAWBalloon realizes the intuitive manipulation of sounds in three dimensions, even for people without knowledge of music.

CCS CONCEPTS

• Human-centered computing;

KEYWORDS

Airborne ultrasound, interaction, music, tangible

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1 INTRODUCTION

The development of music synthesis technology has created a way to enjoy listening to music and actively manipulate it. PerForm [1] is a small, band-shaped musical instrument that we can manage to produce sounds depending on its shape. In an active music application, a user can intuitively manipulate the sound and enjoy the music as if they were the performer. However, it is difficult for an amateur to combine sounds or change pitches to operate a DAW (Digital Audio Workstation).

Several tangible interfaces that manipulate objects as a metaphor for the sound element have been proposed to tackle this issue. However, actuators and air pumps utilized in these interfaces have an

interference problem with the audio modality. Therefore, we focused on ultrasonic levitation and haptic feedback to develop an appropriate interface for DAW. We propose "DAWBalloon," a system that uses ultrasonic levitation arrays to visualize rhythms using floating balloons as a metaphor for musical elements and to combine sounds by manipulating the balloons. DAWBalloon realizes the intuitive manipulation of sounds in three dimensions, even for people without knowledge of music. In this proposal, we first describe an overview of related studies and then explain DAWBalloon.

2 RELATED WORK

Some studies proposed music interfaces representing musical elements with various shapes and materials. SoundForms [2] projects sound waveforms in three dimensions in real-time using a table of pins connected to actuators, allowing the user to manipulate the sound by directly touching the sound waveforms. FabricKeyboard [3] is a deformable music interface that allows users to play music by stretching and shrinking a keyboard made of fabric. MusicBottles [4] is a tangibles user interface using empty transparent bottles. Opening the lid of each bottle plays the sound of a specific instrument, and closing the lid stops the sound. This turns familiar objects into intuitive, manipulatable instruments. Sand Noise Device [5] uses the sandbox environment itself as a musical instrument. Sonic Banana [6] is a device that modulates sound using a bending sensor, enabling music to be manipulated by twisting and bending objects. PocoPoco [7] is a device that plays music by manipulating cylinders arranged in an array on a flat surface. The cylinders have a mechanism that pops them out vertically, allowing the user to tap, hold, and rotate them as input.

Some studies have used spheres as a metaphor for sound elements. For example, Aerial Tunes [8] uses an air-based system to levitate a ball, allowing the sound to vary with the height of the ball. Although the user can manipulate music using the balls, they must maintain a certain distance to avoid the effects of airflow.

Inspired by these previous studies, we propose a system that enables intuitive musical manipulation through levitated objects. The ultrasonic arrays enables simultaneous and fine manipulation by floating objects [9–11], which can tackle the issue of previous work [8] and provide users with various input options.

3 DAWBALLOON: PROJECT MUSICAL ELEMENTS

We propose DAWBalloon, a musical interface that enables intuitive operation by combining balloons assigned to specific rhythms and

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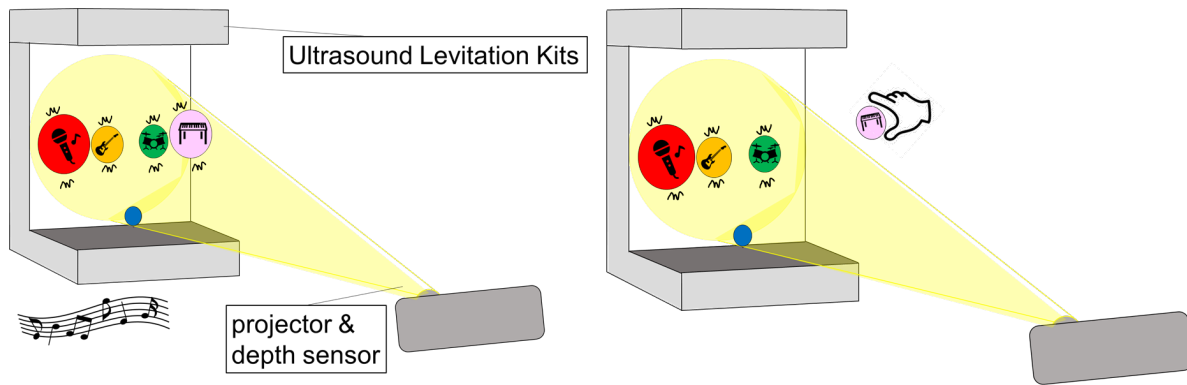


Figure 1: DAWBalloon

sounds. In this section, we describe the components and overview of the proposed system.

3.1 System Components

- **The ultrasonic array** controls the object's levitation, expansion, and contraction
- **Multiple balloons** are utilized by assuming that each balloon corresponds to the sound and rhythm of a specific instrument.
- **A camera with a depth sensor** is used to detect the presence of balloons and the user's hands.
- **A projector** makes the sound element assigned to the each balloons visually intelligible.

3.2 System Overview

DAWBalloon allows users to create music intuitively, regardless of musical knowledge and provides exploratory interaction through sound and visual feedback.

Figure 1 shows an overview of our proposed DAWBalloon. We use ultrasound to control the positions of multiple balloons and project images of playing sound sources onto each balloon. Each balloon has a specific sound source, such as vocals, guitar, and drums. We visually express the sound source by levitating these balloons. In addition, the height position of the balloons presents the volume of each sound source, and the expansion and contraction present the musical rhythm. The interaction of sound manipulation will be realized physically by moving these balloons. Our system plays the specific source of the music when the user floats the balloon and stops when the user removes it. We will also use depth sensors to measure the distance between the user's hand and the balloon and detect the grasping. The depth sensor tracks the position of the user's hand, and when the hand is close to the balloon, the system determines that the user has grasped the balloon and stopped the specific sound source of the music. Then, the system plays it again according to the position when the user places the balloon. In this way, the proposed system could offer intuitive and easy interactions in which the user can manipulate musical elements with the real object.

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